



ECE275 (Sample) Midterm 2 2026

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April 2, 2026

Student Name:

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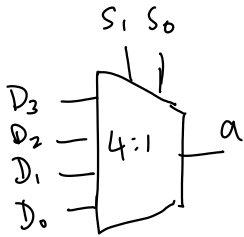
1 Instructions

- There are 6 problems. All problems are required.
- Maximum number of marks is 75.
- Time allowed is 75 minutes.
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8×11 in two-sided cheatsheet is allowed.
- Non-programmable calculators are permitted.
- Please use a pen or heavy pencil to ensure legibility. Colored pens/pencils are recommended for K-map grouping.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

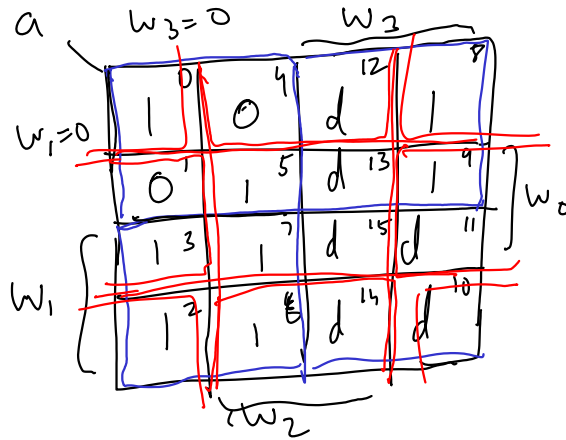
Problem 1. Table 1 shows the truth table for a BCD to 7-segment display. The inputs corresponding to the missing rows in the truth table should be considered as don't care. Implement segment "a" using a 4:1 MUX (multiplexer) and one other gate. (10 marks)

Row	w_3	w_2	w_1	w_0	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Table 1: Truth table for BCD to seven-segment display. The missing combinations of inputs should be considered as don't care.

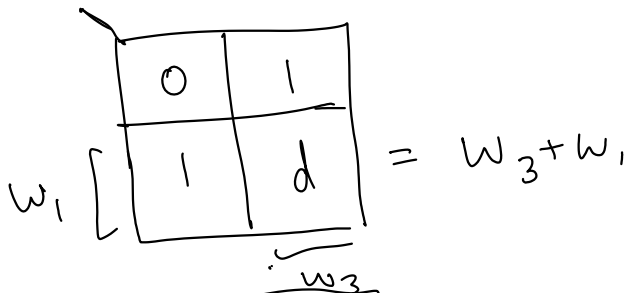


$$S_1 S_0 = w_3 w_1$$

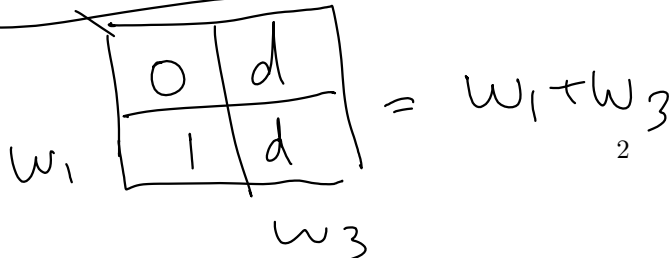


$$S_1 S_0 = w_2 w_0$$

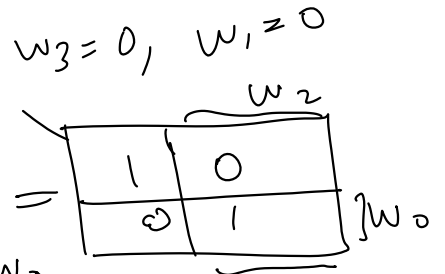
$$w_2 = 0, w_0 = 1$$



$$w_2 = 1, w_0 = 0$$



$$w_2 \oplus w_0 = w_2 \bar{w}_0 + \bar{w}_2 w_0$$



$$S_1 S_0 = w_2 w_0$$

$$w_2 = 0, w_0 = 0$$

$$D_0 = 1$$

$$w_2 = 0, w_0 = 1$$

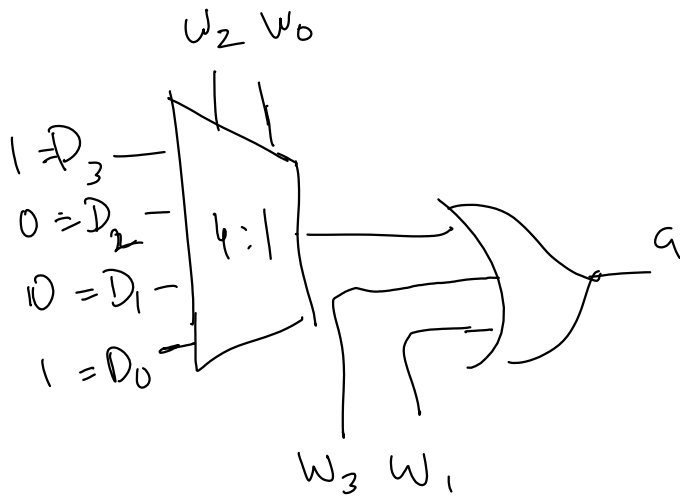
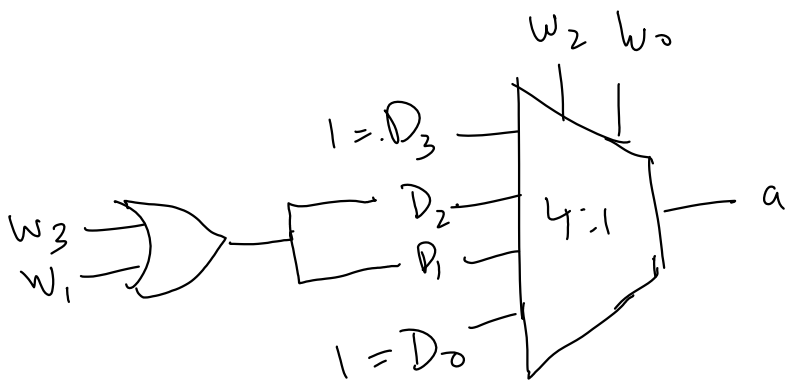
$$D_1 = w_3 + w_1$$

$$w_2 = 1, w_0 = 0$$

$$D_2 = w_3 + w_1$$

$$w_2 = 1, w_0 = 1$$

$$D_3 = 1$$



$$w_2 = 0, w_0 = 1$$

$$w_2 = 1, w_0 = 0$$

Problem 2. *Prove De Morgan laws for sets $\overline{A \cap B} = \bar{A} \cup \bar{B}$, and $\overline{A \cup B} = \bar{A} \cap \bar{B}$ (10 marks).*

Problem 3. Call each statement as true or false for the sets A and B , $A = \{a, e, \{i, o\}, \{u\}\}$ and $B = \{a, i, o, u\}$ (20 marks)

1. $B \in A$
2. $(B - \{i, o\}) \in A$
3. $\{a\} \cup \{\{i, o\}\} \subset A$
4. $\{u\} \in A$
5. $\{\{u\}\} \in A$
6. $B - A = \emptyset$
7. $i \in A \cap B$
8. $\{i, o\} = A \cap B$
9. $\{a\} = A \cap B$
10. $B - A = \{i, o, u\}$

Problem 4. Packages arrive at the stockroom and are delivered on carts to offices and laboratories by student employees. The carts and packages are various sizes and shapes. The students are paid according to the carts used. There are five carts and the pay for their use is

Cart C1: \$2
 Cart C2: \$1
 Cart C3: \$4
 Cart C4: \$2
 Cart C5: \$2

On a particular day, seven packages arrive, and they can be delivered using the five carts as follows:

C1 can be used for packages P1, P3, and P4.

C2 can be used for packages P2, P5, and P6.

C3 can be used for packages P1, P2, P5, P6, and P7.

C4 can be used for packages P3, P6, and P7.

C5 can be used for packages P2 and P4.

The stockroom manager wants the packages delivered at minimum cost. Using minimization techniques described in this class, present a systematic procedure for finding the minimum cost solution. (20 marks)

All packages are delivered

$$= P1 \wedge P2 \wedge P3 \wedge \dots \wedge P7$$

$$= (C1 \vee C3) \wedge (C2 \vee C3 \vee C5) \wedge (C1 \wedge C4)$$

$\wedge (\quad) \wedge \dots$
 Product of sums form

$$= \underbrace{(C1 \wedge C2 \wedge C3)}_{\substack{\text{cost} \\ \$7}} \vee \dots$$

↓ sum of product form

P1: P1 is delivered
 Delivered (P)

C1: Cart 1 is used

Petrick's method

- ① Finding ^{all} prime implicants
 - ② Finding small number of PIs that cover all minterms
- ↑
 packages carts

$$= (C1 \vee C3) \wedge (C2 \vee C3 \vee C5) \wedge (C1 \wedge C4)$$

$\wedge (\quad) \wedge \quad - - -$
 Product of sums form

$$= (C1 + C3)(C2 + C3 + C5)(C1 + C4)$$

$$= (C1 \cdot C2 + C1C3 + C1C5 + \underbrace{C3 \cdot C2 + C3 + C3 \cdot C5}_{\downarrow})(C1 + C4)$$

$$= (C1C2 + C1C3 + C1C5 + C3(\underbrace{C2 + 1 + C5}_{\downarrow})) (C1 + C4)$$

$$= \underline{C1C2} + \underline{C1C3} + C1C5 + C1C3 +$$

$$\underline{C1C2C4} + \underline{C1C3C4} + C3C4$$

$$= C1C2(1 + C4) + C1C3(1 + C4) + \underline{C1C3} + C3C4$$

$$= \boxed{C1C2} + \underline{C1C3} + \underline{C3C4} \quad \leftarrow \text{sum of products}$$

$\underbrace{\quad}_3$
 $\underbrace{\quad}_6$
 $\underbrace{\quad}_6$

Problem 5. Given $A = \{3, 5, 7\}$ $B = \{2, 4, 6, 8, 10, 12\}$ and aRb iff a is a divisor of b , then $R = \{(3, 6), (3, 12), (5, 10)\}$. Is the relation R a valid function? (1 marks)



No because 3 has two
corresponding values 6, 12

x is all Humans

Problem 6. Define an appropriate language and formalize the following sentences using FOL formulas (14 marks).

1. All Students are smart.
2. There exists a student.
3. There exists a smart student.
4. Bill is a student.
5. Bill takes either Analysis or Geometry (but not both).
6. Bill takes Analysis and Geometry.
7. Bill doesn't take Analysis.

Student(x): x is a Student
 Smart(x): x is smart
 Intelect(x, y): x is of intellect y

Predicates
 has variables
 ↓
 have Domain
 | y can be
 { "smart",
 "not smart" }

- ① $\forall x. \text{Student}(x) \wedge \text{Smart}(x)$
 $\forall x. \text{Student}(x) \rightarrow \text{Smart}(x)$

All human are smart students

There exist a student who is smart

- ② $\exists x. \text{Student}(x) \wedge \text{Smart}(x)$

- ③ $\exists x. \text{Student}(x) \rightarrow \text{Smart}(x)$

If there exists a student, then they are smart

- ④ $\exists x. \text{Student}(x)$

- ⑤ Student("Bill")

- ⑥ $\exists x. \text{Student}(x) \wedge (x = \text{"Bill"})$ ✓

~~$\exists x. \text{Student}(\text{"Bill"})$~~

Define constant "Bill"

- ⑦ Takes("Bill", "Analysis") $\wedge$ $\neg$ Takes("Bill", "Geometry")
 Define Takes(x, y): x takes subject y

- ⑧ $\forall (\neg \text{Takes}(\text{"Bill"}, \text{"Analysis"}) \wedge \text{Takes}(\text{"Bill"}, \text{"Geometry"}))$

$$x \oplus y = x \bar{y} + \bar{x} y$$

$$= (x + y) \cdot (\bar{x} \bar{y})$$

$$= (x + y) \cdot (\bar{x} + \bar{y})$$

$$= x \bar{x} + x \bar{y} + y \bar{x} + y \bar{y}$$

$$= x \bar{y} + y \bar{x}$$

⑥ $\text{Takes}(\text{"Bill"}, \text{"Analysis"}) \wedge \text{Takes}(\text{"Bill"}, \text{"Geometry"})$

⑦ $\neg \text{Takes}(\text{"Bill"}, \text{"Analysis"})$

Total marks: 75